

GAH05E 252F - 012

Questions.

01. Compare database, data warehouses and data lakes.

Database	Data warehouses	Data lakes
Store current operational data.	stores historical & integrated data.	stores raw data in its original form.
used for daily transactions. (OLTP)	used for analysis and reporting (OLAP)	stores structured, semi-structured and unstructured data.
ex:- Bank transactions	ex:- monthly sales analysis.	ex:- videos, images, log files.

02. Galle municipal council is interested in building a data warehouse to analyze traffic patterns within the city's road network and reduce congestion during rush hours. Briefly, describe at least two(02) data sources that can assist them in analyzing these patterns and developing strategies. clearly define what to extract?

Data sources :-

01. Police Traffic Records.

* Provides information about traffic accidents and incidents on different roads.

Data to extract: Accident location, date & time, number of accidents, vehicle type, cause of accident.

02. Parking Records.

* Parking records provide information about vehicle parking patterns and busy periods in different area.

Data to extract: Parking location, date & time, vehicle count, vehicle type, parking duration.

03. A tour operator offering trips to Ella in Sri Lanka is interested in understanding visitor preferences and formulating strategies to boost off-peak bookings. Briefly, describe at least two (02) data sources that can assist the operator in analyzing visitor preferences and creating these strategies. Clearly define what to extract?

01. Hotel Records.

Hotel records provide information about tourists' accommodation and their length of stay.

Data to extract: check in/check-out date, number of guests, nationality, length of stay, room/package type.

02. Tourist Attraction Ticket Records

Ticket records provide information about visitor numbers and the popularity of tourist attractions.

Data to extract: Attraction name, visit date & time, number of visitors, ticket type, visitor's nationality.